

6. Academic Performance vs Addiction (Regression Plot)

Description

The Academic Performance vs Addiction regression plot explores the relationship between students' academic performance and their addiction levels. Each point in the graph represents an individual student, while the fitted regression line summarizes the overall trend observed across the dataset. Regression plots are widely used in data analysis to examine whether changes in one variable are associated with changes in another.

Python Code

```
sns.regplot(  
x='academic_performance',  
y='addiction_level',  
data=df,  
scatter_kws={'alpha':0.4})  
plt.title('Academic Performance vs Addiction')  
plt.show()
```

Graph



Interpretation

The scatter points are distributed across the graph without forming a strong linear pattern, and the regression line shows only a weak overall trend. This suggests that academic performance does not exhibit a strong linear relationship with addiction level in the current dataset. The visualization indicates that academic achievement is likely influenced by multiple factors beyond digital addiction, including study habits, sleep quality, motivation, family environment, and psychological well-being. Therefore, while addiction may contribute to academic outcomes, it should not be considered the sole determining factor. More advanced statistical analysis would be required to better understand these complex relationships.